FIITJEE

ALL INDIA TEST SERIES

FULL TEST – IV

JEE (Advanced)-2021

PAPER –2

TEST DATE: 18-01-2021

ANSWERS, HINTS & SOLUTIONS

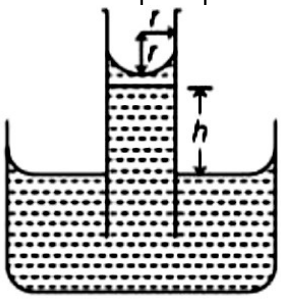
Physics

PART – I

SECTION – A

1. B, C, D

Sol. Volume of liquid upto lower level of meniscus = $\pi r^2 h$

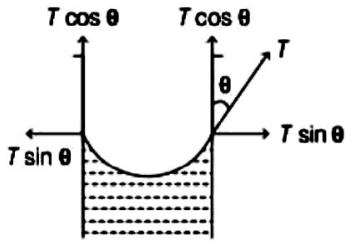


Volume of liquid above meniscus = Volume of cylinder of radius r and height r – Volume of hemisphere of radius $r = \pi r^3 - \frac{2}{3}\pi r^3 = \frac{1}{3}\pi r^3$

So, total volume of liquid in capillary $= \pi r^2 h + \frac{1}{3}\pi r^3 = \pi r^2\left(h + \frac{r}{3}\right)$

For equilibrium, weight of liquid = Upward force of surface tension

$= \pi r^2\left(h + \frac{r}{3}\right) = 2\pi r T\cos\theta$



$$\Rightarrow h = \frac{2T\cos\theta}{r\rho g} - \frac{r}{3}$$

If we neglect $\frac{r}{3}$ in comparison to h, then

$$T = \frac{hr\rho g}{2\cos\theta}$$

or $$T = \frac{r\left(h + \frac{r}{3}\right)\rho g}{2\cos\theta}$$

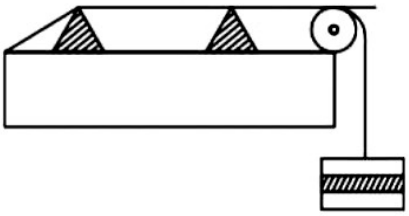
For water and glass, θ is very small

$$\theta \approx 0°$$

$$\therefore \quad T = \frac{r\left(h + \frac{r}{3}\right)\rho g}{2}$$

2. B, C

Sol. For observing resonance paper riders are used and vibrating tuning fork is touched to sonometer box (not with wire). When temperature increases length of prongs of tuning fork increases and thus frequency decreases



Also, $t = \frac{1}{2L}\sqrt{\frac{T}{\mu}}$

$$= \frac{1}{2L}\sqrt{\frac{T}{\left(\frac{\rho D^2 \rho}{4}\right)}}$$

$$= \frac{1}{LD}\sqrt{\frac{T}{\pi\rho}}$$

3. A, D

Sol. Let length l of the cylinder is dipped in liquid, the weight of liquid displaced by the volume of l-part of cylinder

$$\pi r^2 l\rho g = W \qquad \ldots(i)$$

where, ρ = density of the liquid when cylinder is further dipped through a distance x, then volume of displaced [liquid πr2(l + x)]

Therefore, vertically upward force, (i.e. force of buoyance) acting on mass equals $\pi t^2(l + x)\rho g$.
At situation of equilibrium, net force on the mass

$$F = W - \pi r^2(l + x)pg$$

$$= W - \pi r^2 l\rho g - \pi r^2 \rho x g \quad \ldots(ii)$$

From equations (i) and (ii), we get

$$F = -\pi r^2 pgx = -kx$$

$$R = \pi r^2 \rho g$$

The time period, $T = 2\pi\sqrt{\frac{m}{k}}$

$$= 2\pi\sqrt{\frac{m}{\pi r^2 \rho g}}$$

$\therefore$ Frequency of oscillations,

$$v = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{\pi r^2 g\rho}{m}}$$

4. C, D

Sol. If $mg \geq kx - mg$

When spring is taut, the acceleration of each block are same

If $a_2 = 0, T = mg$

$\therefore \quad a_1 = \frac{T + kx - mg}{m} = \frac{kx}{m}$

If mg < kx – mg, string slacks

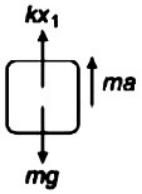
and $a_1 = \frac{kx - mg}{m}$

and $a_2 = g$

5. A, C

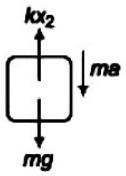
Sol. For upward moving elevator,

$$kx_1 - mg = ma \quad \ldots(i)$$



For downward moving elevator,

$$mg - kx_2 = ma \quad \ldots(ii)$$

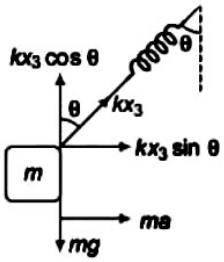


Dividing Eq. (i) by Eq. (ii), we get

$$\frac{g + a}{g - a} = \frac{x_1}{x_2} = \frac{4\sqrt{2}}{3\sqrt{2}}$$

$\therefore \quad a = \dfrac{g}{7}$

For horizontal motion of elevator, spring will be inclined at some angle as given in figure.



$kx_3 \cos\theta = mg \qquad \ldots(iii)$

$kx_3 \sin\theta = ma \qquad \ldots(iv)$

Squaring and adding Eqs. (iii) and (iv), we get

$(kx_3)^2 = m^2(g^2 + a^2)$

$x_3 = \dfrac{m}{k}\sqrt{g^2 + a^2} \qquad \ldots(v)$

From Eqs. (i) and (v), we get

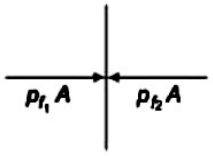
$\dfrac{x_3}{x_1} = \sqrt{\dfrac{g^2 + a^2}{g + a}}$

$x_3 = \left(\dfrac{\sqrt{g^2 + a^2}}{g + a}\right)$

Hence, x_3 = 5 mm

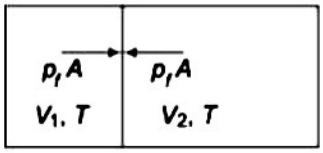
6. A, B, C, D

Sol. At final equilibrium, force must be equal from two side, so



$p_{I_1}A = p_{I_2}A \Rightarrow p_{I_1} = p_{I_2}$

In final condition, final temperature will also be same, because separator is conducting.



Conservation of moles in I and II components gives

$\dfrac{pV}{RT} = \dfrac{p_t(V_1)}{RT} \qquad \ldots(i)$

$\dfrac{2p(2V)}{RT} = \dfrac{p_t V_2}{RT} \qquad \ldots(ii)$

$V_1 = \dfrac{pV}{p_t}$

$$V_2 = \frac{4pV}{p_t} \qquad \ldots(iii)$$

$$V_1 + V_2 = 3V$$

$$\frac{pV}{p_t} + \frac{4pV}{p_t} = 3V$$

$$\frac{pV}{p_t} \times 5 = 3V \Rightarrow p_t = \frac{5\rho}{3}$$

So, $\quad V_1 = \frac{3V}{5}, V_2 = \frac{12V}{5}$

SECTION – B

7. 7

Sol. Find the network done for the process, then find heat supplied. By this you can find the value x with the help of efficiency formula.

$$\text{Efficiency} = \frac{\text{Total work done}}{\text{Total heat supplied}}$$

$$W_{AB} = p(\Delta V) = nR\Delta T \quad \text{(isochoric process)}$$

$$= 2 \times \frac{25}{3} \times 300 = 5000 \text{ J}$$

$$W_{BC} = nRT\ln\left(\frac{T_f}{t_i}\right) = nRT\ln\left(\frac{p_i}{p_t}\right) \text{ (isothermal process)}$$

$$= 2 \times R \times 600\ln\left(\frac{2\times10^5}{1\times10^5}\right)$$

= 1200 R ln (2)

$$W_{CA} = \frac{nr.\Delta T}{2-\gamma} = \frac{-2R\times300}{1-\frac{1}{2}} \qquad \begin{bmatrix} pT = k \\ p^2V = K \\ p \times V^{1/2} = K \end{bmatrix}$$

= –1200 R

$$W = 600R + 1200R\ln(2) - 1200R$$

$$W = 600R[2\ln(2) - 1]$$

Heat supplied,

$$Q_{AB} = nC_p\Delta T = 2 \times \frac{5R}{2} \times 300 = 1500R$$

$$Q_{BC} = W_{BC} = 1200R\ln(2)$$

$$\eta = \frac{600R[2\ln(2)-1]}{1500R + 1200R\ln(2)} = \frac{12\ln(2)-6}{15+12\ln(2)}$$

$$\eta = 1 - \frac{21}{15+12\ln(2)}$$

$$= 1 - \frac{3\times7}{15+12\ln(2)}$$

$\therefore$ x = 7

8. 3

Sol. In this problem, concept of friction as well as the concept of torque will be used. For the equilibrium, resultant force will be zero. We well also balance the torque for the equilibrium.

$$\Sigma F = 0$$

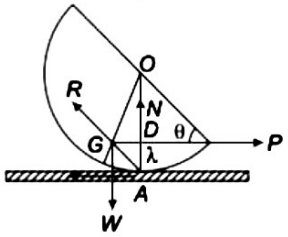
$$\Sigma \tau = 0$$

Under equilibrium,

$$\Sigma F_H = 0 = P - \mu N = 0 \Rightarrow P = \mu N$$

$$\Sigma F_V = 0 = N - W = 0 \Rightarrow N = W$$

$$P = \mu W$$



Moment about contact point A,

$$P(AD) = W(GD)$$

$$P(OA - OD) = W(GD)$$

$$\sin\theta = \frac{P}{P + \frac{4W}{3\pi}} \qquad \ldots(ii)$$

Substituting, P = μw from Eq. (i)

$$\sin\theta = \frac{\mu W}{\mu W + \frac{4W}{3\pi}}$$

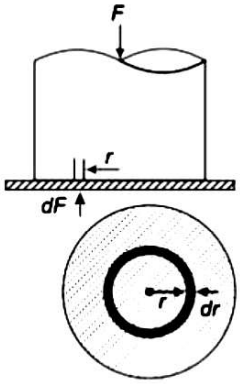
$$\sin\theta = \frac{3\mu\pi}{3\mu\pi + 4}$$

$$\Rightarrow \quad \theta = \sin^{-1}\frac{3\mu\pi}{3\mu\pi + 4}$$

$$7\sin\theta = 3$$

9. 3

Sol. Consider a small element of radius r and width dr.



Normal reaction, $dF = \frac{F}{\frac{\pi}{4}D^2} \times 2\pi r dr$

Frictional force, $dF_1 = \mu dF$

Frictional moment, $dM = rdF_t = r\mu dF$

Total moment,

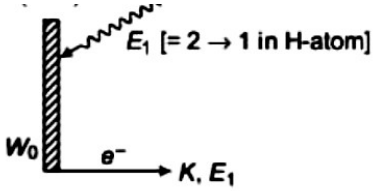
$$M = \int dM = \int_O^{D/2} r\mu \times \frac{4F}{\pi D^2} \times 2\pi r dr$$

$$= \frac{4F}{\pi D^2} \times \mu \times 2\pi \times \frac{(D/2)^3}{3}$$

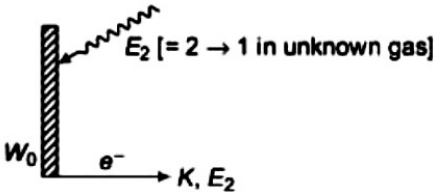
$$M = \frac{\mu FD}{3} \Rightarrow N = 3$$

10. 2

Sol.



We have $E_1 = W_0 + KE_1$ ...(i)



$E_2 = W_0 + KE_2$ ...(ii)

Also, $\lambda_{e^-} \propto \frac{1}{\sqrt{KE}}$

$$\frac{KE_2}{KE_1} = \left(\frac{\lambda_1}{\lambda_2}\right)^2 = 6.1 \qquad \left[\because \lambda_2 = \frac{1}{\sqrt{6.1}}\lambda_1\right]$$

$E_1 = 13.61 \times 1^2\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$ = 10.2 eV ...(iv)

Now, ionisation energy of H-atoms,

= – E_1 = 13.6 eV energy of first line in Lyman series $= 13.6Z^2\left(\frac{1}{1^2} - \frac{1}{2^2}\right)$eV

$= 10.2\ Z^2$eV

and energy of series limit of Balmer series $= 13.6 \times Z^2\left(\frac{1}{2^2} - \frac{1}{\infty^2}\right)$eV $= 3.4Z^2$eV

Given, $13.6 \times 2 = 10.2\ Z^2 - 3.4\ Z^2$

$\Rightarrow$ Z = 2 $\Rightarrow$ He^+ ion

11. 1

Sol. For constructive interference, intensity (I) is maximum,

When cosϕ = +1

The path difference at point P,

$\Delta x = (SS_2 - SS_1) + (S_2P - S_1P)$

$$= \frac{dy}{D_1} + \frac{d(d/2)}{D_2}$$

For constructive interference,

$$\Delta x = \frac{dy}{D_1} + \frac{d^2}{2D_2} = n\lambda$$

$$\frac{(10^{-3})(0.5\sin\pi t)\times 10^{-3}}{1} + \frac{(10^{-3})^2}{2\times 2} = n\lambda$$

$$0.5\sin\left(\frac{\pi}{6}\right)t\times 10^{-6} + 0.25\times 10^{-6}$$

$$= (5000\times 10^{-10})n = 0.5\times 10^{-6}n$$

$$\sin\left(\frac{\pi}{6}\right)t = \frac{0.5n - 0.25}{0.5}$$

For the minimum value of t, n = 1

$$\sin\left(\frac{\pi}{6}\right)t = \frac{1}{2} \Rightarrow \left(\frac{\pi}{6}\right)t = \frac{\pi}{6}$$

or t = 1s

12. 5

Sol. $$\Delta W_{AB} = nC_P\Delta T = \frac{\gamma}{\gamma-1}nR\Delta T = \frac{\gamma}{\gamma-1}$$

$$[3p_0V_0 - p_0V_0] = 2pV_0 \times \frac{\gamma}{\gamma-1}$$

$$\Delta Q_{AC} = \Delta U + \Delta W$$

$$= \frac{nR}{\gamma-1}\Delta T + \frac{1}{2}\times 3V_0[p_0 + 4p_0]$$

$$= \frac{[16p_0V_0 - p_0V_0]}{\gamma-1} + \frac{15p_0V_0}{2}$$

$$56 = 2p_0V_0 \times \frac{\gamma}{\gamma-1}$$

$$360 = 15p_0V_0\left[\frac{\gamma+1}{2(\gamma-1)}\right]$$

$$\frac{360}{56} = \frac{15}{4}\cdot\frac{(\gamma+1)}{\gamma}$$

$$12\gamma = 7\gamma + 7, \gamma = \frac{7}{5} = 1 + \frac{2}{f}$$

$\Rightarrow$ t = 5

SECTION – C

13. 00002.00

Sol. For the given area total force will be u dA

When dA is the area, then electrical force per unit area

$$= \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0\left(\frac{\sigma}{2\varepsilon_0}\right)^2 = \frac{\sigma^2}{2ge_0}$$

Projection area $= \pi r^2$

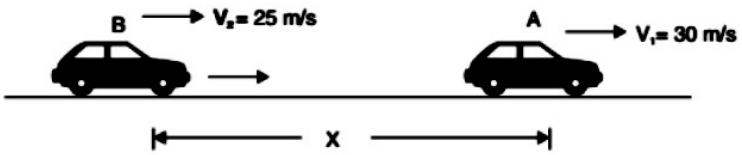
$\Rightarrow$ Net electrical force $= \left(\frac{\sigma^2}{2\varepsilon_0}\right)(\pi R^2)$

In equilibrium, this force should be equal to the applied force,

$$F = \frac{\sigma^2 R^2}{2\varepsilon_0} = \frac{4\varepsilon_0}{2\varepsilon_0} = 2$$

14. 00023.33

Sol.



Let x = distance between the car at the instant first bullet is fired (sat at time t = 0)

Speed of bullet, relative to ground is

$V_b = u + V_2 = 305+25 = 330$ m/s

Velocity of bullet, relative to car A is

$V_{bA} = 330 – 30 = 330$ m/s

$\therefore$ Time when the first bullet hits the car a is

$$t_1 = \frac{X}{V_{bA}} = \frac{X}{300} \qquad \ldots(i)$$

Distance between the car when the last bullet is fired (at time t = T0 = 20 s) is

$$X^1 = X + V_{AB}T_0 = X + 5 \times 20 = X + 100$$

Time when the last bullet hits the car A is

$$t_2 = T_0 + \frac{X}{V_{bA}}$$

$$= 20 + \frac{X+100}{300} \qquad \ldots(ii)$$

The interval $\Delta t = t_2 – t_1$

$$= 20 + \frac{100}{30} = 23.33\,\text{sec}.$$

15. 00000.75

Sol. Consider the tube + liquid inside it + Block as our system.

For equilibrium

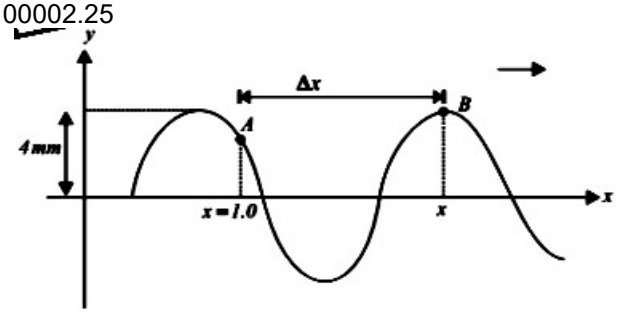
Weight = Buoyancy

$$2\rho_\omega vg + \pi r^2 . \frac{L}{2}\rho_\omega g = \left[v + \pi r^2 \frac{L}{2}\right]\rho_\omega g$$

Solving, v = 0.75 m^3

16. 00002.25

Sol.



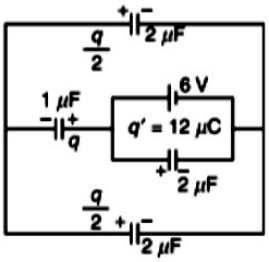
We have to find x co-ordinate of particle B. Phase difference between A and B is

$$\phi = \frac{5\pi}{3} \Rightarrow \frac{2\pi}{\lambda}(\Delta x) = \frac{5\pi}{3}$$

$$\Delta x = \frac{5 \times 1.5}{6} = 1.25\text{m}; \ \therefore x_B = 2.25\ \text{m}$$

17. 00016.80

Sol. The potential difference across 1 μF and 2 μF (uppermost) capacitor must sum up to

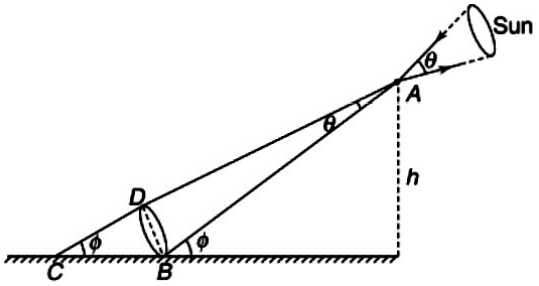


$$\frac{q}{1} + \frac{q/2}{2} = 6 \Rightarrow q = \frac{24}{5}\mu C$$

$\therefore$ Answer is $12 + \frac{24}{5} = 16.8\mu C$

18. 00003.44

Sol. The cone of rays passing through the hole at A produce an elliptical spot on the floor. The circular base having diameter BD will get projected on the floor as an ellipse.



CB = major axis = 12 cm

DB = Minor axis = 6 cm

$$\text{DB} \simeq \text{AB}(\theta) = \frac{h\theta}{\sin\phi} \Rightarrow h = (6\text{ cm})\frac{\sin\phi}{\theta}$$

But $\sin\phi = \frac{\text{DB}}{\text{CB}} = \frac{6}{12} = \frac{1}{2}$

$\therefore$ $h = \frac{(6\text{ cm})}{2 \times \theta}$

$$= \frac{3\text{ cm} \times 180}{0.5 \times \pi}\left[0.5^\circ = \frac{0.5 \times \pi}{180}\text{radian}\right]$$

Chemistry

PART – II

SECTION – A

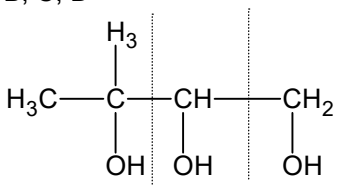
19. A, B, C, D
Sol. Factual

20. A, B, D
Sol. There will be post in each possible geometrical isomer thus no-optical isomers.

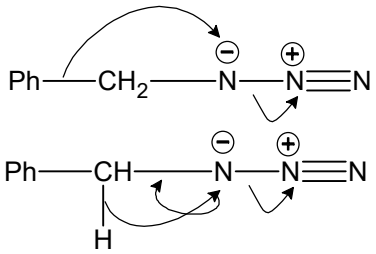
21. A, C, D
Sol. Factual

22. B, C
Sol. POS in B
POS in C after even
Interchanges.

23. B, C, D
Sol.



24. B, C
Sol.



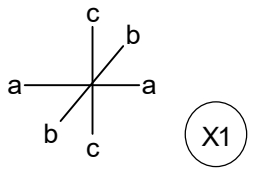
SECTION – B

25. 3
Sol. $Q = H_2O_2$
$R = O_2$

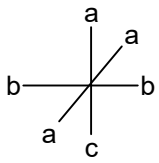
26. 0
Sol. $[Si_1O_{1+2}]^{2-}$

27. 6

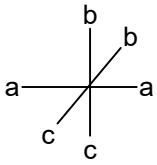
Sol.



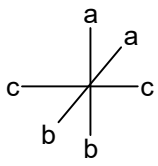
X1



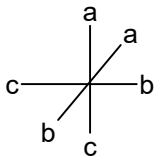
X1



X1



X1



X2

Pair of enantiomers

28. 6

Sol. $C_xH_{12} + (x+3)O_2 \rightarrow xCO_2 + 6H_2O$

At 400 K H_2O is gaseous.

29. 5

Sol. Compound 1, 3, 4, 5, 6.

30. 3

Sol. Compound 2, 4, 5.

SECTION – C

31. 00033.00

Sol. X_2O_3

$X_1^{3+}X_a^{2+}X_{1-a}^{5+}O_3$

$3 + 2a + 5(1-a) - 6 = 0$

[balancing charges]

32. 00044.00

Sol. $\Delta H = \Delta U + P_2V_2 - P_1V_1$

33. 00600.00

Sol. $\underset{2}{PCl_5} \rightleftharpoons \underset{2}{PCl_3} + \underset{2}{Cl_2}$

After 'x' moles Cl_2 is added.

$2+y \quad 2-y \quad 2+x-y$

$2 + x - y + 2 - y + 2 + y = 2 \times (6)$

$\Rightarrow x - y = 6$

$2+y \quad 2-y \quad 8$

$$\frac{\frac{2}{V} \times \frac{2}{V}}{\frac{2}{V}} = \frac{8/2V}{\left(\frac{2-y}{2v}\right) \times \left(\frac{2+y}{2V}\right)}$$

34. 00287.00

Sol. $\frac{K}{S} = \lambda^0_{Ag^+} + X^o_{cl^-}$

Convert mol/l to g/l

35. 00054.00

Sol. Let molar mass = M

$$\frac{\frac{6}{M}}{\frac{180}{18}+\frac{6}{M}} \times P^o = 20$$

$$\frac{\frac{6}{M}}{11+\frac{6}{M}} \times P^o = 20.02$$

36. 00020.00

Sol. $\Delta H - \Delta U = \Delta ng \times RT$

Mathematics

PART – III

SECTION – A

37. A, B, C

Sol. $I_1 = \int_0^{\pi/2} \cos\left(\pi\cos^2 x\right)dx = \int_0^{\pi/2} \cos\left(\pi\sin^2 x\right)dx$

$2I_2 = \int_0^{\pi/2} 2\cos\left(\frac{\pi}{2}\right)\cos\left(\frac{\pi}{2}\cos 2x\right) = 0$

$I_1 = 0$

$I_2 = \int_0^{\pi/2} \cos\left(\pi\left(1-\cos 2x\right)\right)dx = -\int_0^{\pi/2} \cos\left(\pi\cos 2x\right)dx$

$= -\frac{1}{2}\int_0^{\pi} \cos\left(\pi\cos t\right)dt \qquad$ put $2x = t$

$= -I_3$

$\Rightarrow I_2 + I_3 = 0$

38. B, D

Sol. The line y= mx + c mater a chord of length $2\sqrt{\frac{a^2\left(1+m^2\right)-c^2}{1+m^2}}$ from circle $x^2 + y^2 = a^2$

39. A, B, D

Sol. $y = \log_7\left(\log_7 x\right) \Rightarrow \frac{dy}{dx} = \frac{1}{x\log e^x \log_e 7}$

40. A, B, D

Sol. $Y - y = -\frac{1}{m}\left(X - x\right) \Rightarrow X + mY - \left(x + my\right) = 0$

Perpendicular distance from origin $= \left|\frac{x+my}{\sqrt{1+m^2}}\right| = y$

$\Rightarrow x^2 + 2xym = y^2 \qquad \Rightarrow \frac{dy}{dx} = \frac{y^2 - x^2}{2xy}$

Also $x.2y.\frac{dy}{dx} - x^2 = y^2$

Put $y^2 = t$, $\frac{dt}{dx} - \frac{1}{x}.t = -x$

41. A, C

Sol. $I = \int x.\ln\left(1+x^2\right)dx = \ln\left(1+x^2\right).\frac{x^2}{2} - \int \frac{1}{1+x^2}2x.\frac{x^2}{2}dx$

$= \frac{x^2}{2}\ln\left(1+x^2\right) - \int \frac{x^3.dx}{x^2+1}$

$$I = \left(\frac{x^2+1}{2}\right)\ln\left(1+x^2\right) - \left(\frac{x^2+1}{2}\right) + c$$

42. B, D

Sol. $\frac{x^2}{4} - \frac{9}{4}x.\frac{1}{3} = 1 \qquad x^2 - 3x - 4 = 0 \qquad x = 4, -1$

Points $(4, \pm 3)$

$\therefore$ Required equation(s) of tangents are $x \pm y = 1$

SECTION – B

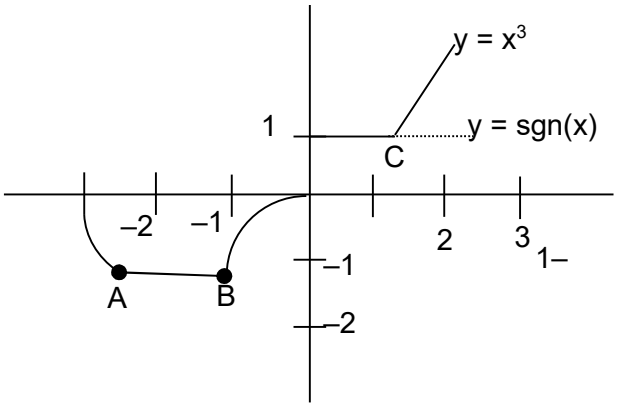
43. 8

Sol. $x^2 + x + 1 = 0 \Rightarrow x + \frac{1}{x} = -1$

$$\therefore \sum_{r=1}^{5}\left(x^r + \frac{1}{x^r}\right)^2 = 8$$

44. 5

Sol. Let at A, B, C f(x) is continuous but not differentiable at A, B, C. Right hand derivative is $-\infty$ and A + is left hard derivative is $+\infty$.



$\therefore$ 5 points

45. 1

Sol. Equation of line through P(1, -2, 3) and parallel to given line is $\frac{x-1}{2} = \frac{y+3}{3} = \frac{z-3}{-6} = \lambda$

General point online $Q(2\lambda + 1, 3\lambda - 2, -6\lambda + 3)$ if then point lie on plane, $\lambda = \frac{1}{7}$

$$\therefore Q\left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$$

$$\therefore PQ = \sqrt{\frac{4}{49} + \frac{9}{49} + \frac{36}{49}} = 1$$

46. 7

Sol. $9^x - 3^x + 1 = \left(3^2 - \frac{1}{2}\right)^2 + \frac{3}{4} \geq \frac{3}{4}$

$x \in (-\infty, 1) \frac{3}{4} \le 9^x - 3^x + 1 \le 7$

Range of f(x) = {0, 1, 2, 3, 4, 5, 6}

47. 6

Sol. If let A denotes the event that target is hit when x shells are fired at point I.

$P(E_1) = \frac{8}{9}, \quad P(E_2) = \frac{1}{9}$

$P\left(\frac{A}{E_1}\right) = 1 - \left(\frac{1}{2}\right)^x$ ard $P\left(\frac{A}{E_2}\right) = 1\left(\frac{1}{2}\right)^{21-x}$

$P(A) = \frac{8}{9}\left(1 - \left(\frac{1}{2}\right)^x\right) + \frac{1}{9}\left(1 - \left(\frac{1}{2}\right)^{21-x}\right)$

$\frac{d(A)}{dx} = 0 \quad \Rightarrow x = 12 \quad \frac{d^2P(A)}{dx} < 0$

$\therefore \alpha = 12 \quad \frac{\alpha}{2} = 6$

48. 2

Sol. $\alpha = \int_0^{\pi} (x + \sin x)\,dx - \int_0^{\pi} x\,dx = 2$ square unit

SECTION – C

49. 00400.00

Sol. $Am = \frac{100 + n}{2}$, $Gm = 10\sqrt{n}$, $Hm = \frac{200n}{100 + n}$

For Am to be integer, n is even

For GM to be integer n should be perfect square

$\Rightarrow$ $n = 4k^2$

$\therefore Hm = \frac{200k^2}{25 + k^2} \quad \Rightarrow k^2 = 125, 100, 75, 25$

$\therefore$ $k^2 = 100$ only satisfy.

$x = 4$ $kx^2 = 400$

50. 00208.00

Sol. TTT, A, E, R, S

3 like, 1 distinct $= 4_{C_1} \times \frac{4!}{3!} = 16$

2 like, 2 distinct $= 4_{C_2} \times 1 \times \frac{4!}{2!} = 72$

All for different $= 5_{C_4} \times 4! = 120$

$\therefore$ Total number of words = 16 + 72 + 120 = 208

51. 00000.00

Sol. $z = \begin{vmatrix} -5 & 3+5i & \frac{3}{2}-4i \\ 3-5i & 8 & 4+5i \\ \frac{3}{2}+4i & 4-5i & 9 \end{vmatrix}$

$z = \bar{z} \quad \therefore z \in$ purel, real

$\therefore$ imaginer part of z = 0

52. 00004.00

Sol. $\left[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}\right] = \left[\vec{a}\ \vec{b}\ \vec{c}\right]^2 = \begin{bmatrix} \vec{a}.\vec{a} & \vec{a}.\vec{b} & \vec{a}.\vec{c} \\ \vec{b}.\vec{a} & \vec{b}.\vec{b} & \vec{b}.\vec{c} \\ \vec{c}.\vec{a} & \vec{c}.\vec{b} & \vec{c}.\vec{c} \end{bmatrix} = 32$

$\therefore \ \left[\vec{a}\vec{b}\vec{c}\right]\frac{1}{12} = 4$

53. 00004.25

Sol. An P(2, 0) lie on curve – 89 + 4b – 2c + 5 = 0 ....(1)

$Q(0, 5)\frac{dy}{dx} = 3ax^2 + 2bx + c$

$At(0, 5)\frac{dy}{dx} = 3 \Rightarrow c = 3 \qquad ...(2)$

$\frac{dy}{dx}at\ (-2, 0) = 0 \qquad \Rightarrow 12a - 4b + 3 = 0$

$\therefore a = -\frac{1}{2},\ b = -\frac{3}{4},\ c = 3$

$\left|c - a - b\right| = \left|3 + \frac{2}{4} + \frac{3}{4}\right| = \frac{17}{4} = 4.25$

54. 00004.00

Sol. $\cos 2\theta = -\sin\theta = \cos\left(\frac{\pi}{2} + 3\theta\right)$

$2\theta = 2n\pi \pm \left(\frac{\pi}{2} + 3\theta\right),\ n \in I$

$for + ve\ sign\ \ \theta = -\left(2n\pi + \frac{\pi}{2}\right)$

For –ve sign $\theta = \frac{2n\pi}{5} - \frac{\pi}{10}$

$\therefore \theta = \frac{3\pi}{10}, \frac{7\pi}{10}, \frac{11\pi}{10}, \frac{15\pi}{10}$

Hence four values